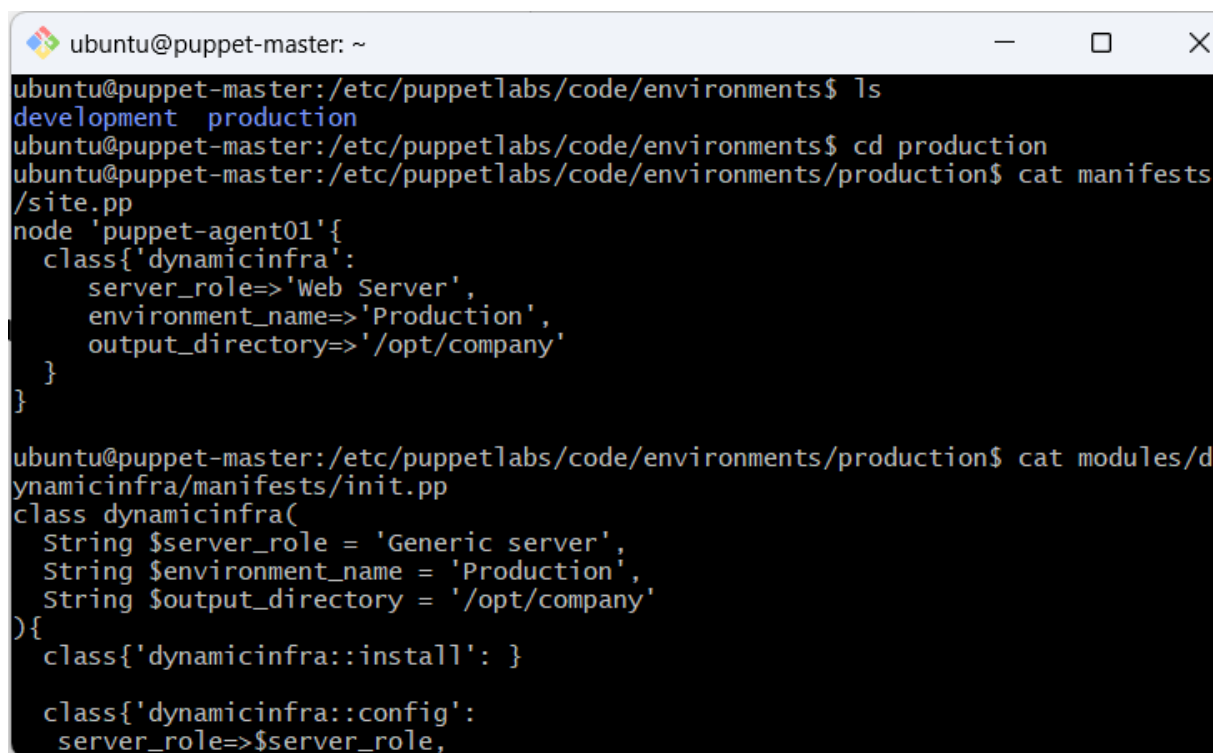


Complete Puppet Mastery Part-4

Practical 13 - Enterprise Multi-Environment Management Using Puppet Environments(Standalone Flagship Practical)

Note: Have used production env for Puppet-agent01 and development for Puppet-agent02

Master:

A terminal window titled 'ubuntu@puppet-master: ~' with standard window controls. The terminal shows the following commands and output:

```
ubuntu@puppet-master:/etc/puppetlabs/code/environments$ ls
development  production
ubuntu@puppet-master:/etc/puppetlabs/code/environments$ cd production
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ cat manifests/site.pp
node 'puppet-agent01'{
  class{'dynamicinfra':
    server_role=>'Web Server',
    environment_name=>'Production',
    output_directory=>'/opt/company'
  }
}

ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ cat modules/dynamicinfra/manifests/init.pp
class dynamicinfra(
  String $server_role = 'Generic server',
  String $environment_name = 'Production',
  String $output_directory = '/opt/company'
){
  class{'dynamicinfra::install': }

  class{'dynamicinfra::config':
    server_role=>$server_role,
```

```
ubuntu@puppet-master: ~  
{  
  class{'dynamicinfra::install': }  
  
  class{'dynamicinfra::config':  
    server_role=>$server_role,  
    environment_name=>$environment_name,  
    output_directory=>$output_directory  
  }  
  
  class{'dynamicinfra::service': }  
  
  Class['dynamicinfra::install']->Class['dynamicinfra::config']  
  ->Class['dynamicinfra::service']  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ cat modules/d  
ynamicinfra/manifests/config.pp  
class dynamicinfra::config(  
  String $server_role,  
  String $environment_name,  
  String $output_directory  
)  
{  
  
  $service_name = $facts['os']['family'] ? {  
    'Debian'=>'apache2',  
    'RedHat'=>'httpd',  
    default=>'apache2'  
  }  
  
  file{$output_directory:  
    ensure=>directory,  
    owner=>'root',  
    group=>'root',  
    mode=>'0755'  
  }  
  
  file{"${output_directory}/system-info.txt":  
    ensure=>file,  
    owner=>'root',  
    group=>'root',  
    mode=>'0644',  
  
    content=>epp(  
      'dynamicinfra/system-info.epp',  
      {  
        'server_role'=>$server_role,  
        'environment_name'=>$environment_name,  
        'output_directory'=>$output_directory  
      }  
    )  
  }  
}
```

```
ubuntu@puppet-master: ~  
$service_name = $facts['os']['family'] ? {  
  'Debian'=>'apache2',  
  'RedHat'=>'httpd',  
  default=>'apache2'  
}  
  
file{$output_directory:  
  ensure=>directory,  
  owner=>'root',  
  group=>'root',  
  mode=>'0755'  
}  
  
file{"${output_directory}/system-info.txt":  
  ensure=>file,  
  owner=>'root',  
  group=>'root',  
  mode=>'0644',  
  
  content=>epp(  
    'dynamicinfra/system-info.epp',  
    {  
      'server_role'=>$server_role,  
      'environment_name'=>$environment_name,  
      'output_directory'=>$output_directory  
    }  
  )  
}
```

```
ubuntu@puppet-master: ~
    'environment_name'=>$environment_name
  }
),
notify=>Service[$service_name]
}

if $facts['os']['family'] == 'Debian' {

  file{"${output_directory}/os-info.txt":
    ensure=>file,
    owner=>'root',
    group=>'root',
    mode=>'0644',
    content=>"This node belongs to the Debian family.
    Package Manager: APT\n"
  }
}
elseif $facts['os']['family'] == 'RedHat' {

  file{"${output_directory}/os-info.txt":
    ensure=>file,
    owner=>'root',
    group=>'root',
    mode=>'0644',
```

```
ubuntu@puppet-master: ~
    group=>'root',
    mode=>'0644',
    content=>"This node belongs to the Redhat family.\nPackage Manager:
    DNF/YUM\n"
  }
}
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ cat modules/d
ynamicinfra/manifests/service.pp
class dynamicinfra::service{

  if $facts['os']['family'] == 'Debian' {

    service{'apache2':
      ensure=>'running',
      enable=>true
    }
  }

  elseif $facts['os']['family'] == 'RedHat'{

    service{'httpd':
      ensure=>'running',
      enable=>true
```

```
ubuntu@puppet-master: ~  
}   
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ cat modules/d  
ynamicinfra/manifests/install.pp  
class dynamicinfra::install{  
  
    package{'tree':  
        ensure=>installed  
    }  
  
    package{'curl':  
        ensure=>installed  
    }  
  
    if $facts['os']['family'] == 'Debian' {  
        package{'apache2':  
            ensure=>installed  
        }  
    }  
    elsif $facts['os']['family'] == 'RedHat'{  
        package{'httpd':  
            ensure=>installed  
        }  
    }  
}
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development  
ubuntu@puppet-master:~$ cd /etc/puppetlabs/code/environments  
ubuntu@puppet-master:/etc/puppetlabs/code/environments$ cd development  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/development$ cat manifest  
s/site.pp  
node 'puppet-agent02'{  
    class{'dynamicinfra':  
        server_role=>'Application server',  
        environment_name=>'Development',  
        output_directory=>'/opt/company',  
        experimental_features=>true  
    }  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/development$ cat modules/  
dynamicinfra/manifests/init.pp  
class dynamicinfra(  
    String $server_role = 'Generic server',  
    String $environment_name = 'Production',  
    String $output_directory = '/opt/company',  
    Boolean $experimental_features = false  
) {  
    class{'dynamicinfra::install': }  
  
    class{'dynamicinfra::config':  
        server_role=>$server_role,  

```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development
class{'dynamicinfra::config':
  server_role=>$server_role,
  environment_name=>$environment_name,
  output_directory=>$output_directory,
  experimental_features=>$experimental_features
}

class{'dynamicinfra::service': }

Class['dynamicinfra::install']->Class['dynamicinfra::config']
->Class['dynamicinfra::service']
}
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development$ cat modules/
dynamicinfra/manifests/config.pp
class dynamicinfra::config(
  String $server_role,
  String $environment_name,
  String $output_directory,
  Boolean $experimental_features
){
  $service_name = $facts['os']['family'] ? {
    'Debian'=>'apache2',
    'RedHat'=>'httpd',
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development
    'RedHat'=>'httpd',
    default=>'apache2'
  }

  file{$output_directory:
    ensure=>directory,
    owner=>'root',
    group=>'root',
    mode=>'0755'
  }

  file{"${output_directory}/system-info.txt":
    ensure=>file,
    owner=>'root',
    group=>'root',
    mode=>'0644',

    content=>epp(
      'dynamicinfra/system-info.epp',
      {
        'server_role'=>$server_role,
        'environment_name'=>$environment_name
      }
    ),
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development
    'server_role'=>$server_role,
    'environment_name'=>$environment_name
  }
),
notify=>Service[$service_name]
}

if $experimental_features{

  file { "${output_directory}/development-warning.txt":

    ensure => file,
    owner   => 'root',
    group   => 'root',
    mode     => '0644',

    content => epp(
      'dynamicinfra/development-warning.epp',
      {
        'server_role'      => $server_role,
        'environment_name' => $environment_name,
      }
    ),
  },
}
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development

  notify => Service[$service_name],
}
}

if $facts['os']['family'] == 'Debian' {

  file{"${output_directory}/os-info.txt":
    ensure=>file,
    owner=>'root',
    group=>'root',
    mode=>'0644',
    content=>"This node belongs to the Debian family.
    Package Manager: APT\n"
  }
}

elseif $facts['os']['family'] == 'RedHat' {

  file{"${output_directory}/os-info.txt":
    ensure=>file,
    owner=>'root',
    group=>'root',
    mode=>'0644',
  }
}
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development
    ensure=>file,
    owner=>'root',
    group=>'root',
    mode=>'0644',
    content=>"This node belongs to the Redhat family.\nPackage Manager:
      DNF/YUM\n"
  }
}
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/development$ cat modules/
dynamicinfra/manifests/install.pp
class dynamicinfra::install{

  package{'tree':
    ensure=>installed
  }

  package{'curl':
    ensure=>installed
  }

  if $facts['os']['family'] == 'Debian' {

    package{'apache2':
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development
  }
}
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/development$ cat modules/
dynamicinfra/manifests/service.pp
class dynamicinfra::service{

  if $facts['os']['family'] == 'Debian' {

    service{'apache2':
      ensure=>'running',
      enable=>true
    }
  }

  elsif $facts['os']['family'] == 'RedHat'{

    service{'httpd':
      ensure=>'running',
      enable=>true
    }
  }
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/development$ |
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development/...  
icinfra/templates$ sudo cat development-warning.epp  
=====   
                DEVELOPMENT ENVIRONMENT REPORT   
=====   
  
WARNING  
  
This node belongs to the Development Puppet Environment.  
  
Changes deployed here are intended for testing before  
promotion to Production.  
  
Server Role : <%= $server_role %>  
  
Environment : <%= $environment_name %>  
  
Hostname : <%= $facts['networking']['hostname'] %>  
  
Primary IP : <%= $facts['networking']['ip'] %>  
  
Operating System : <%= $facts['os']['name'] %>  
  
Generated By : Puppet Development Environment and Vishnu's Passion
```

Agent-01:

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Refreshing CA certificate  
Info: CA certificate is unmodified, using existing CA certificate  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.14.115)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1785940898'  
Notice: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]/con  
tent:  
--- /opt/company/system-info.txt      2026-08-04 12:18:59.323048678 +0000  
+++ /tmp/puppet-file20260805-1846-eo6hb5      2026-08-05 14:41:39.479455533 +0  
000  
@@ -17,7 +17,7 @@  
Architecture: amd64  
  
Processors: 2  
-Memory: 914.01 MiB  
+Memory: 914.00 MiB  
  
Puppet Version: 8.10.0
```

```
ubuntu@puppet-agent01: ~  
Notice: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]/con  
tent: content changed '{sha256}3cae9473383acedf08ff0eea520a378e06c14cab7e2828378  
5ce0605e73f8e73' to '{sha256}b65c0566132bb39034d5ad23cbc96e621a095be7e0c2553401c  
b51477da5a086'  
Info: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]: Sche  
duling refresh of Service[apache2]  
Notice: /Stage[main]/Dynamicinfra::Service/Service[apache2]: Triggered 'refresh'  
from 1 event  
Notice: Applied catalog in 0.61 seconds  
ubuntu@puppet-agent01:~$ ls /opt/company  
os-info.txt system-info.txt  
ubuntu@puppet-agent01:~$ cat /opt/company/os-info.txt  
This node belongs to the Debian family.  
Package Manager: APT  
ubuntu@puppet-agent01:~$ cat /opt/company/system-info.txt  
-----  
Enterprise Infrastructure Report  
-----  
  
Server Role: Web Server  
Environment: Production
```

```
ubuntu@puppet-agent01: ~  
Environment: Production  
  
Hostname: puppet-agent01  
FQDN: puppet-agent01.ap-south-1.compute.internal  
Primary IP: 172.31.7.84  
  
Operating System: Ubuntu  
OS Release: 22.04  
OS Family: Debian  
  
Kernel: Linux  
Architecture: amd64  
  
Processors: 2  
Memory: 914.00 MiB  
  
Puppet Version: 8.10.0  
  
Generated By: Puppet Master and Vishnu's Passion  
  
-----  
ubuntu@puppet-agent01:~$
```

Agent-02:

```
ubuntu@ip-172-31-14-217: ~  
ubuntu@ip-172-31-14-217:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Using environment 'development'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.14.115)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent02  
Info: Applying configuration version '1785940306'  
Notice: /Stage[main]/Dynamicinfra::Config/File[/opt/company/development-warning.txt]/ensure: defined content as '{sha256}da73f6188783c7bc5063881faea40f7aad08aa433478a0dc95e7e3612b3e1348'  
Info: /Stage[main]/Dynamicinfra::Config/File[/opt/company/development-warning.txt]: Scheduling refresh of Service[apache2]  
Notice: /Stage[main]/Dynamicinfra::Service/Service[apache2]: Triggered 'refresh' from 1 event  
Notice: Applied catalog in 0.56 seconds  
ubuntu@ip-172-31-14-217:~$ ls /opt/company  
development-warning.txt  os-info.txt  system-info.txt  
ubuntu@ip-172-31-14-217:~$ cat os-info.txt  
cat: os-info.txt: No such file or directory  
ubuntu@ip-172-31-14-217:~$ cat /opt/company/os-info.txt  
This node belongs to the Debian family.  
Package Manager: APT  
ubuntu@ip-172-31-14-217:~$ cat /opt/company/system-info.txt
```

```
ubuntu@ip-172-31-14-217: ~  
ubuntu@ip-172-31-14-217:~$ cat /opt/company/system-info.txt  
-----  
Enterprise Infrastructure Report  
-----  
  
Server Role: Application server  
Environment: Development  
  
Hostname: puppet-agent02  
FQDN: puppet-agent02.ap-south-1.compute.internal  
Primary IP: 172.31.14.217  
  
Operating System: Ubuntu  
OS Release: 22.04  
OS Family: Debian  
  
Kernel: Linux  
Architecture: amd64  
  
Processors: 2  
Memory: 914.01 MiB  
  
Puppet Version: 8.10.0
```

```
ubuntu@ip-172-31-14-217: ~
Puppet Version: 8.10.0

Generated By: Puppet Master and Vishnu's Passion

-----
ubuntu@ip-172-31-14-217:~$ cat /opt/company/development-warning.txt
=====
DEVELOPMENT ENVIRONMENT REPORT
=====

WARNING

This node belongs to the Development Puppet Environment.

Changes deployed here are intended for testing before
promotion to Production.

Server Role : Application server

Environment : Development

Hostname : puppet-agent02
```

```
ubuntu@ip-172-31-14-217: ~
GNU nano 6.2 /etc/puppetlabs/puppet/puppet.conf
# This file can be used to override the default puppet settings.
# See the following links for more details on what settings are available:
# - https://puppet.com/docs/puppet/latest/config_important_settings.html
# - https://puppet.com/docs/puppet/latest/config_about_settings.html
# - https://puppet.com/docs/puppet/latest/config_file_main.html
# - https://puppet.com/docs/puppet/latest/configuration.html
[main]
server = puppet-master
[agent]
certname = puppet-agent02
environment = development

[ Read 12 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut      ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste    ^J Justify  ^_ Go To Line
```

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Vishnu@0407 (36899587946)

Vishnu@0407

EC2 > Instances

Instances (3/3) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

EC2

Dashboard

AWS Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

1

☒	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	F
☒	puppet-master	i-0c0b95ffc889341a7	Running	t3.small	3/3 checks passed	ap-south-1b	ec2-43-204-233-8.ap-s...	4
☒	puppet-agent01	i-093cc3c71fb0dd8d8	Running	t3.micro	3/3 checks passed	ap-south-1b	ec2-13-126-24-116.ap-...	1
☒	puppet-agent02	i-0f1d92d83de5c9f5a	Running	t3.micro	3/3 checks passed	ap-south-1b	ec2-65-0-21-155.ap-so...	6

3 instances selected

Alarm recommendations

CPU utilization (%)

Percent

1.39%

0.695%

0%

14.00

14.30

Network in (bytes)

Bytes

33.32k

16.66k

0

14.00

14.30

Network out (bytes)

Bytes

30.67k

15.33k

0

14.00

14.30

Network packets in (count)

Count

311

155.5

0

14.00

14.30

CloudShell

Agent Toolkit for AWS

Feedback

Console Mobile App

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